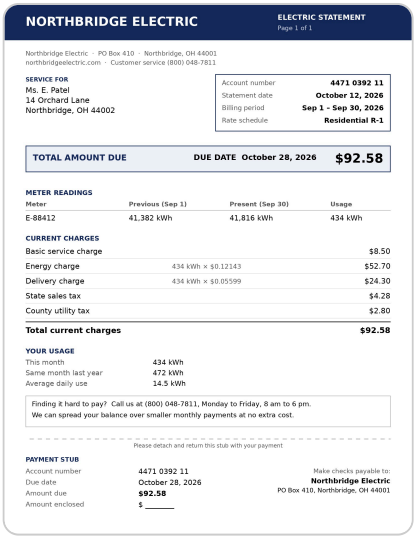
BigPrint TV — how to try it

A guide for judges and anyone else. Five minutes with a Vega virtual device; fifteen seconds without one. No AWS account, no sign-in. Backend live to 20 November 2026.

What it is. Photograph a letter, a bill or a note with any phone. It appears on the television in very large print, with the figures that matter pinned at the top, and one press reads it aloud. Built for people with age-related sight loss, on the screen they already own.



NORTHBRIDGE ELECTRIC

ELECTRIC STATEMENT

Page 1 of 1

Northbridge Electric · PO Box 410 · Northbridge, OH 44001
northbridgeelectric.com · Customer service (800) 048-7811

SERVICE FOR
Ms. E. Patel
14 Orchard Lane
Northbridge, OH 44002

Account number	4471 0392 11
Statement date	October 12, 2026
Billing period	Sep 1 – Sep 30, 2026
Rate schedule	Residential R-1

TOTAL AMOUNT DUE — DUE DATE October 28, 2026 — $92.58

METER READINGS

Meter	Previous (Sep 1)	Present (Sep 30)	Usage
E-88412	41,382 kWh	41,816 kWh	434 kWh

CURRENT CHARGES

Basic service charge		$8.50
Energy charge	434 kWh × $0.12143	$52.70
Delivery charge	434 kWh × $0.05599	$24.30
State sales tax		$4.28
County utility tax		$2.80
Total current charges		**$92.58**

YOUR USAGE

This month	434 kWh
Same month last year	472 kWh
Average daily use	14.5 kWh

Finding it hard to pay? Call us at (800) 048-7811, Monday to Friday, 8 am to 6 pm. We can spread your balance over smaller monthly payments at no extra cost.

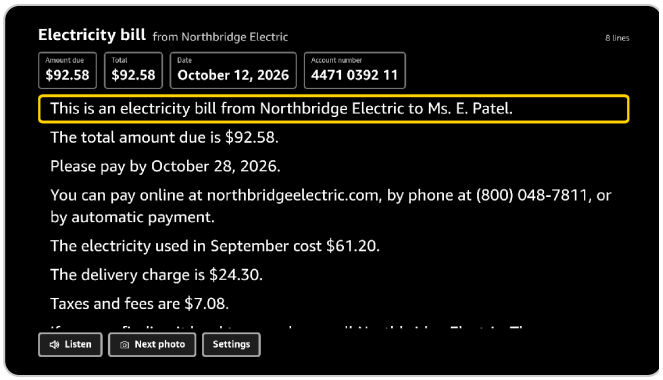
Please detach and return this stub with your payment

PAYMENT STUB

Account number	4471 0392 11
Due date	October 28, 2026
Amount due	$92.58
Amount enclosed	$ _____

Make checks payable to: Northbridge Electric
PO Box 410, Northbridge, OH 44001

The bill you will photograph: `samples/us-bill.jpg`.



What the TV shows about ten seconds later.

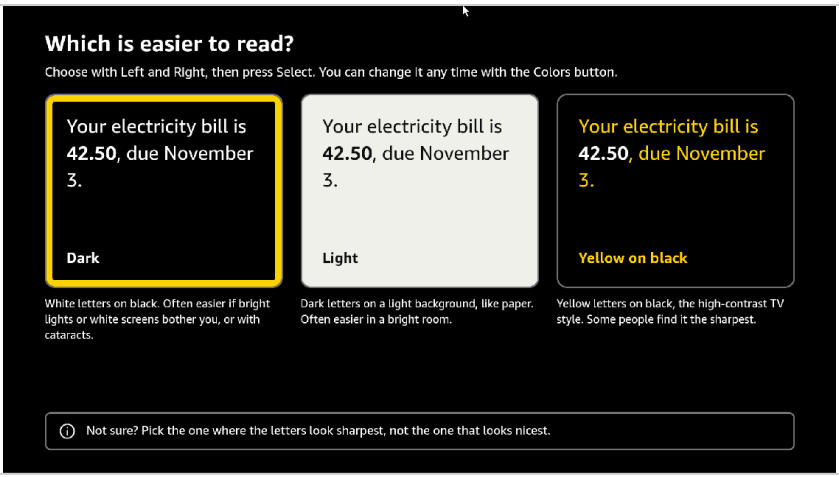
1. Install

You need the Vega SDK with its virtual device, or a Fire TV running Vega OS in developer mode. Download the package for your target from **github.com/rukhsarai/bigprint-tv/releases/latest** (also in the Devpost upload): `x86_64` for the virtual device, `aarch64` for a Fire TV.

```
vega virtual-device start          # skip on a real Fire TV
vega install-app bigprint-tv-0.1.0_x86_64.vpkg
vega run-app com.bigprint.tv.main
```

2. The first run

1. The app opens on a **choose-a-look** screen: the same sentence in dark, light, and yellow on black. Move with the D-pad and press Select on the one you can read best. Two people with the same eye condition often choose differently, which is why the app asks by showing, not by naming.

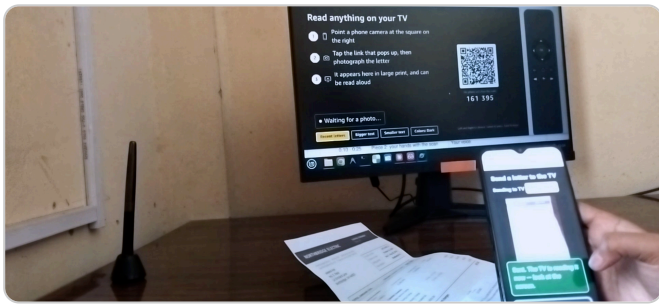


First run: pick a look by reading a real sentence.

2 The TV then shows a **QR code and a six-digit code**. Scan the QR with any phone's camera; a web page with one button opens. Nothing to install on the phone.



The pairing screen.



Any phone scans it.

3 **Photograph the bill.** Open `samples/us-bill.jpg` on a laptop screen and photograph that, or print it. Hold the phone square to the page and fill the frame. Press the one button.

4 The TV says *Reading your letter*, and about ten seconds later the page is on screen: what it is and who it is from, the key figures pinned at the top, and the whole letter in large print below. Move line by line with the D-pad; the focused line has a thick yellow ring.

Check the TV against the paper. It should show amount due **$92.58**, due date **October 28, 2026**, statement date October 12, 2026, account **4471 0392 11**. Every one of those came from OCR. If the TV ever shows a figure the paper does not, that is a bug worth reporting.

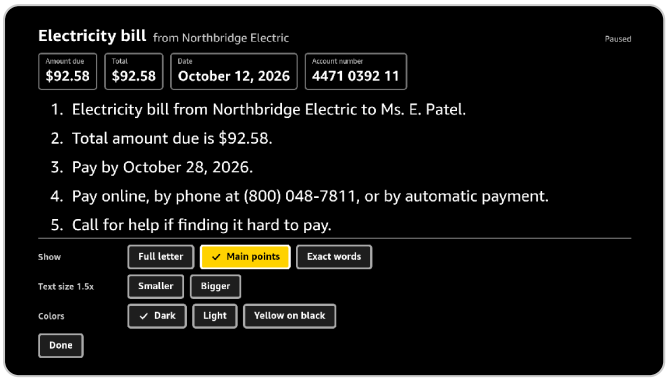
3. Things to try

Listen

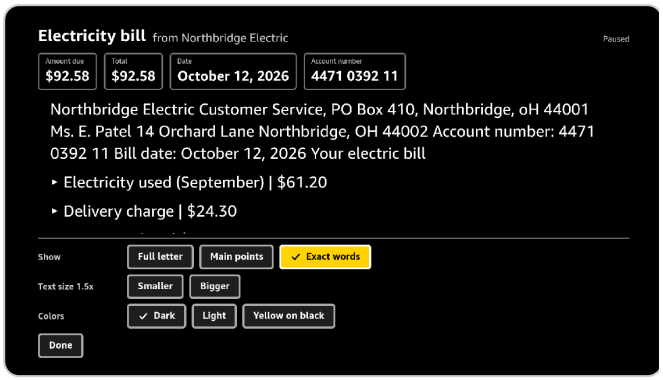
Press **Listen** (the first button at the bottom). The whole page is read aloud in a natural voice; the button becomes **Pause**. The voice is made only now, on request, which is why the text arrived first.

Three ways to read it

Settings → Show switches between the whole letter in plain English, just the **main points**, and the **exact words** as printed, with table rows kept as rows.



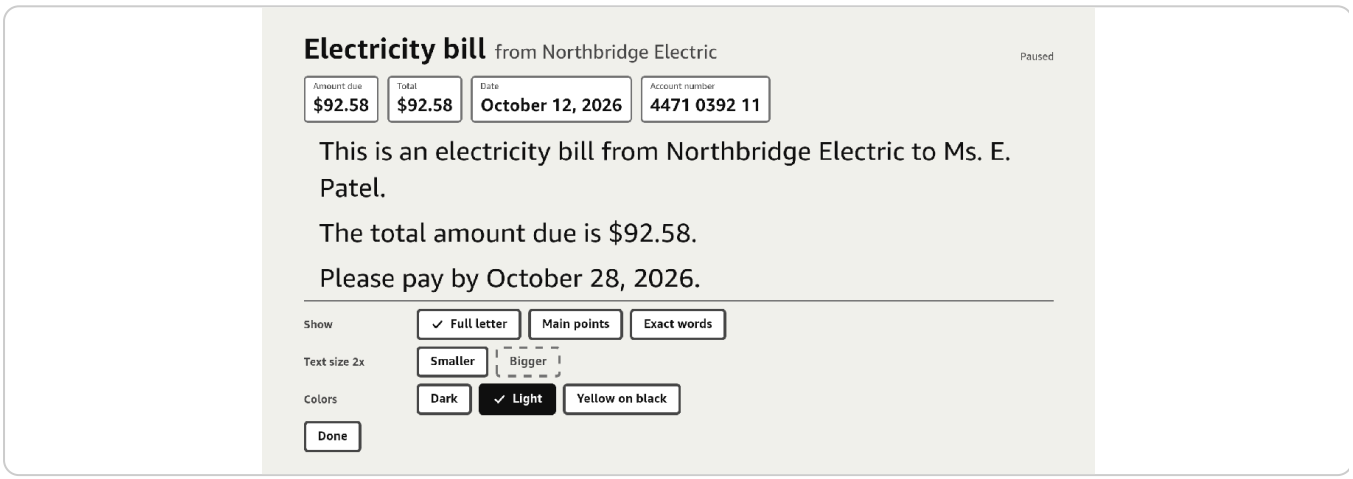
Main points.



Exact words.

Text size and looks

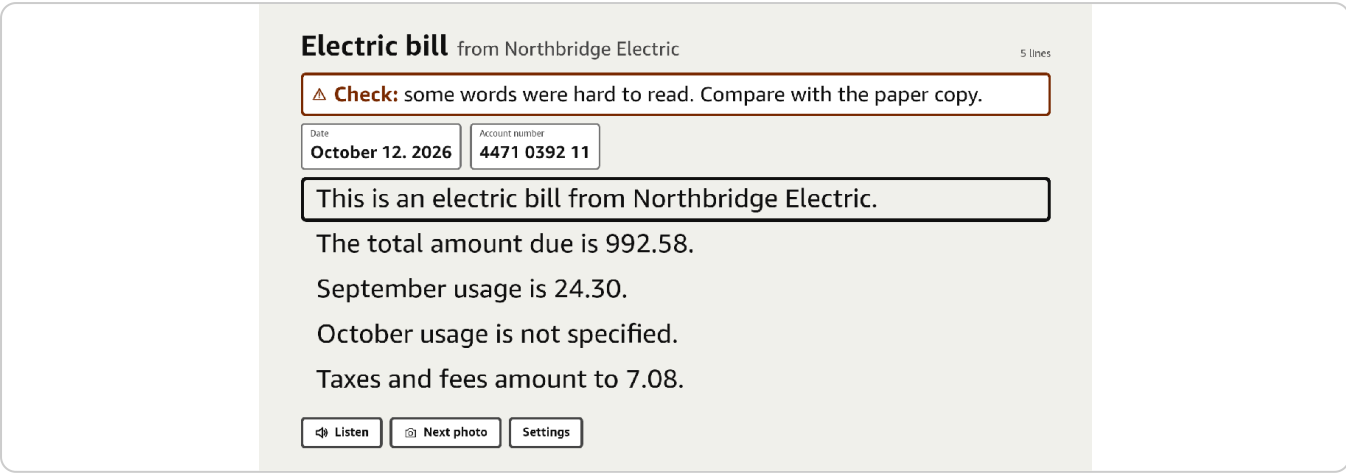
Settings → Text size goes up to 2×; the pinned figures never scroll away. **Settings → Look** switches dark, light, and yellow on black. Every colour pair in all three passes WCAG 7:1, checked in the test suite.



The light look at 2× text: the pinned figures stay put while the letter grows.

A page that is hard to read

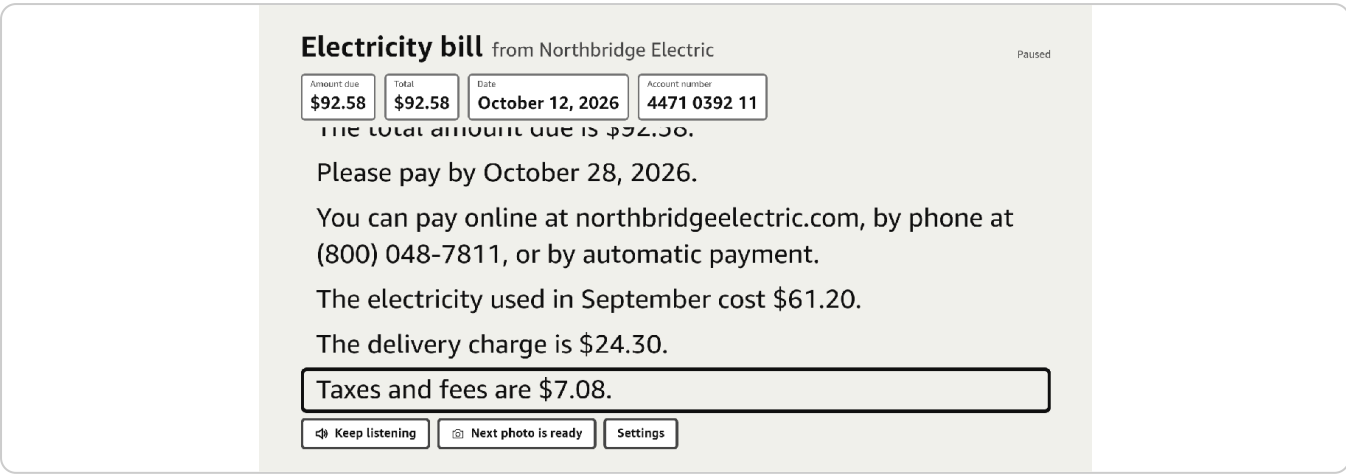
Photograph `samples/us-bill-blurry.jpg`, a deliberately poor photo of the same bill. The TV says *Check: some words were hard to read* and shows what it is unsure about, instead of guessing.



The honest answer to a blurry page: a Check line first, and only the figures it is sure of.

Next photo, and Recent letters

Next photo takes you back to the pairing screen for the next page; the phone page is still open, so it is one more tap. Press **Back** on the remote from the letter to see **Recent letters**, held on the TV for a day, then deleted. Back never exits the app by accident.



While you read, the next page can already be waiting: the middle button says so.

4. Without a device

From a terminal with `curl` and Python, in the repository:

```
backend/tools/round_trip.sh              # live backend, end to end, ~15 s
backend/tools/round_trip.sh samples/us-bill.jpg   # the same with the video's bill
```

It opens a pairing session, uploads a photo (by default it draws `samples/uk-water-bill.jpg`), waits for the text, asks for the voice, and prints the key facts the pipeline extracted and the time each step took. That is the whole product minus the screen.

```
npm ci && npm test                           # app: 157 tests
python3 -m unittest discover backend/tests   # backend: 37 tests
```

5. What is going on underneath

Step	Service	Rule
Read the photo, with layout and tables	Amazon Textract	A bill's rows stay rows
Classify the page	Amazon Nova Lite	Bill, letter, receipt, certificate
Pull the money figures	Textract AnalyzeExpense	Numbers come from OCR, never from a model
Second look at the photo	Amazon Nova Pro	May veto a figure; can never add one
Rewrite into spoken sentences	Amazon Nova Pro	Any sentence with a number not on the page is dropped
Voice, on request	Amazon Polly	Made only when someone presses Listen

Two Lambda functions, one S3 bucket, an API Gateway in front. Nothing runs between letters. Measured on the sample photos: 17 key facts shown, 0 wrong. About three cents a page, five with the voice.

Before you photograph a real letter. Photos go to AWS (us-east-1) to be read. The photo, the text and the audio are deleted automatically after one day, logs never contain the text of a page, and the public endpoint cannot call any AI service directly. It is still a hackathon deployment, so please use the samples, or a page you do not mind a demo server seeing for a day. There is a cap of 200 pages a day across everyone and 80 per network address; if the TV says the service is busy, that is why.

6. Known limits

Not yet run on a physical Fire TV with VoiceView, the screen reader; the accessibility markup is right by construction and by test, not yet by ear. At the largest text sizes the pairing-screen instructions can overflow on some viewports. The rewrite occasionally adds "More items are listed." when nothing was left out; the numeric guard means it can never carry a wrong figure.

Code, packages and the eleven-entry developer friction log: **github.com/rukhsarai/bigprint-tv**. MIT licence. Questions: open an issue there.